Problem set 6
Due Thursday March 5 at 11:59pm

Exercise 6.1 (8 points).

(1) Prove that the interval $(0,1)$ in $\mathbb{R}$ is not complete.

(2) Suppose X is a set equipped with the discrete metric. Is X complete?

Exercise 6.2 (8 points; Rudin 3.24). Suppose (p_n) and (q_n) are Cauchy sequences in a metric space X. Prove that the sequence $(d(p_n, q_n))$ in $\mathbb{R}$ has a limit.

Exercise 6.3 (8 points). Let (s_n) be the sequence defined by

$$s_1 = 0 \quad s_{2n} = s_{2n-1} + 1 \quad s_{2n+1} = \frac{s_{2n}}{2}$$

Find the upper and lower limits of (s_n) (a.k.a lim sup and lim inf.) Prove your answers.

Exercise 6.4 (8 points; Rudin 3.21). Recall that for a subset E of a metric space, we define $\operatorname{diam} E = \sup\{d(p,q) \mid p, q \in E\}$. Suppose (E_n) is a sequence of closed, nonempty sets in a complete metric space X, with $E_n \supset E_{n+1}$, and $\lim_{n\to\infty} \operatorname{diam} E_n = 0$. Prove that $\cap_{n=1}^{\infty} E_n$ consists of exactly one point.

Exercise 6.5 (not for credit). Suppose $E \subset X$. We say E is *sequentially closed* if every sequence (p_n) in E which is convergent in X is actually convergent in E. Prove that E is closed if and only if it is sequentially closed.

Exercise 6.6 (not for credit). Suppose (p_{2n}) and (p_{2n+1}) both converge to the same limit q. Prove that (p_n) converges to q.

Exercise 6.7 (not for credit; challenge). This exercise explains the name of lim sup. Show that for any sequence of real numbers (s_n),

$$\limsup_{n\to\infty} s_n = \lim_{n\to\infty} \sup\{s_m | m \geq n\}$$

Exercise 6.8 (not for credit; challenge). Let r be a an irrational positive real number. Let

$$s_n = nr - \lfloor nr \rfloor$$

where $\lfloor a \rfloor$ denotes the greatest integer less than the real number a. Show that every point in the interval $[0,1]$ is a subsequential limit of (s_n). (This is a bit involved, at least the way I did it. In dynamics, one would phrase roughly this result as "an irrational rotation generates a minimal action on the circle.")